



UNIVERSITY
of **HULL**

Computer Science

BSc / MEng Software Engineering

October 2026

Software Engineering

Shaded modules are offered as options on your programme, all other modules are compulsory.

Level 4 (Certificate Stage)

Trimester 1	Trimester 2
Programming Portfolio (40cr. core)	
Algorithms and Data Structures	Databases and the Cloud
Professional Development	Computational Thinking

Common first year allows for easy transfer between the available programmes.

Level 5 (Diploma Stage)

Trimester 1	Trimester 2
Design, Develop, Deploy (40cr. core)	
Artificial Intelligence	Distributed Systems Programming
Foundations of Cybersecurity and Secure Development	Advanced Programming

Post-Diploma Stage (optional)

Trimester 1	Trimester 2
Year in Industry/Year Abroad	

Level 6 (Honours Stage)

Trimester 1	Trimester 2
Honours Stage Project (40 credits)	
Communicating and Teaching Computing (40 credits)	
Secure Digital Infrastructure	Safety Critical Systems
Digital Disruption and Innovation	Generative and Collaborative AI

Level 7 (Masters Stage) (optional)

Trimester 1	Trimester 2
Commercial Development Practice (40 credits)	
Dependable Reactive Real-Time Systems	Cutting Edge Concepts
Component Based Software Engineering Practice	Trustworthy Computing

Please note that the content in the following pages is subject to change, as our curriculum evolves to match the rapidly changing Computer Science landscape.

Level 4

Programming Portfolio

Core tools: C#

Whether you've never written a line of code or you've been tinkering for years, this module meets you where you are. You'll learn to think like a programmer — breaking problems down, writing clean logic, and building working software in C#. By the end of the year you'll have moved from your very first program to designing with classes, objects and inheritance.

You'll explore:

- Core programming building blocks — variables, loops, functions and data types
- Searching, sorting and working with arrays, stacks and queues
- Object-oriented programming — classes, inheritance and encapsulation
- Reading, writing and managing code like a professional, using source control

Algorithms and Data Structures

Every great piece of software starts with a great idea for how to solve the problem — that's what this module teaches you to find. You'll step away from the screen and learn to think in code before you write it, exploring the algorithms and data structures that separate software that merely works from software that works brilliantly.

You'll explore:

- Pseudocode and state machines for planning before you program
- Core data structures, and how to choose the right one for the job
- Sorting and searching algorithms, and how to measure their efficiency
- Design thinking that carries into every other module on your degree

Databases and the Cloud

Core tools: C#, Microsoft SQL Server

Modern software rarely stores its data on just one machine anymore — this module shows you how databases and the cloud work together to make that possible. You'll learn how cloud computing and virtualisation actually work, then design and build real databases, from relational systems through to NoSQL, before bringing the two together to build a working application backed by a database hosted in the cloud.

You'll explore:

- Designing and implementing relational and NoSQL databases
- Virtualisation, containers, and the cloud service models behind them — IaaS, PaaS and SaaS
- Connecting a real application to a cloud-hosted database

- Cloud security, reliability and disaster recover

Computational Thinking

Core tools: programming platforms and mathematical software

Computing is bigger than any one programming language — this module is about how to think as a computer scientist. Drawing on a genuinely global history of innovation, you'll build the mathematical and logical foundations that everything else on your degree rests on.

You'll explore:

- Abstraction and computational methods
- The maths behind computing — algebra, number systems and statistics
- Logic, Boolean algebra and set theory
- Vectors and matrices, and their uses in graphics, AI and data
- Practical aspects of how computers handle different number types and the way that mathematical libraries and tools can help to do the actual calculations

Professional Development

A computer science degree is about more than code — this module helps you build the career to go with it. You'll explore the ethical, legal, social and sustainability challenges facing the sector, and the professional skills that set graduates apart. A hands-on team project puts those skills into practice straight away, while you start building a portfolio that showcases your work.

You'll explore:

- Sector challenges — ethics, diversity, sustainability and cybersecurity
- Communication, research and self-presentation skills
- Real teamwork on a programme-specific group project
- Building your own professional portfolio

Level 5

Design, Develop, Deploy

Core tools: *GitHub*, *GitHub Actions*

This is where you learn to build software the way the industry actually builds it — from first idea to live deployment. As the backbone module of your second year, you'll work through the entire software development lifecycle, solo and as part of a team, picking up the methodologies and tools employers expect you to know.

You'll explore:

- Agile, iterative and waterfall development approaches
- Gathering and modelling real-world requirements
- Test-driven and behaviour-driven development
- Version control, CI/CD and cloud deployment with GitHub Actions

Advanced Programming

Core tools: C++

C++ powers everything from games engines to spacecraft — this module shows you why, from the inside out. You'll push beyond the basics of object-oriented programming and get close to the machine, exploring what's really happening between your code and the processor.

You'll explore:

- Templates and the Standard Template Library
- Composition and inheritance in advanced data structures
- Pointers, memory management and the “gang of three”
- Assembly language and debugging at the processor level

Artificial Intelligence

Learn to build the kind of intelligent systems that are reshaping every industry and understand the ethics behind them. You'll get a practical grounding in how AI actually works, from the history of the field to the algorithms powering it today, building your own solutions to real problem-solving challenges along the way.

You'll explore:

- The history and ethics and trustworthiness of artificial intelligence
- Pathfinding and problem-solving search algorithms
- Supervised and unsupervised learning, and neural networks
- Reasoning under uncertainty with Bayesian methods, and an introduction to reinforcement learning
- A brief introduction of fairness, explainability and AI Safety

Distributed Systems Programming

Core tools: .NET Core

Modern software rarely runs on just one machine — learn to build systems that work seamlessly across many. You'll design and build real distributed applications, navigating the unique challenges of concurrency, scalability, performance, security and scale that come with connecting devices and sharing workloads across a network.

You'll explore:

- Architectures and strategies for distributed computing
- Concurrency, security, performance and reliability
- Key distributed design patterns and best practice
- Deploying layered, scalable distributed applications

Foundations of Cybersecurity and Secure Development

Cybersecurity is at the heart of today's digital world. This module provides a broad introduction to the principles, technologies, and practices that protect systems, networks, software, and data from cyber threats. Students will explore how security is integrated into the design and development of digital solutions and gain an understanding of the technical, organisational, and ethical aspects of cybersecurity. The module establishes the essential knowledge required for more advanced study in cybersecurity throughout the degree.

You'll explore:

- Fundamental cybersecurity concepts and security principles
- Cyber threats, vulnerabilities, and risk management approaches
- Secure system and software development practices
- Data protection fundamentals
- Governance, ethics, and legal aspects of cybersecurity
- Emerging trends and technologies in cybersecurity

Year in Industry/Year Abroad

Take everything you've learned so far and put it to work — in a real workplace or a different country altogether. Spend a year gaining genuine professional experience on an industry placement, or broaden your horizons studying at an overseas institution. Either route builds transferable, employability-boosting skills in a credit-bearing year that counts towards your degree.

You'll explore:

- A year of real workplace experience, tailored to your placement
- Or a year studying abroad, guided by department staff to match your goals

Level 6

Honours Stage Project

This is your chance to lead a project entirely your own — from first idea to final report. You'll choose your own topic and see it through from planning to delivery, supported by videos, live sessions and workshops covering everything you need to succeed.

You'll explore:

- Project planning and literature review
- Experimental design, testing and evaluation
- Research ethics and data protection
- Professional report writing and presentation

Communicating and Teaching Computing

Think you understand computer science? Try explaining it to someone else — that's the real test. Spend real time in schools or colleges as a computing ambassador, supported by workshops on teaching practice and the National Curriculum, and discover how explaining your subject deepens your own understanding of it.

You'll explore:

- Classroom management and lesson planning
- How computer science is taught in schools today
- Reflective, experience-based learning from your own placement
- Effective communication skills – a critical life skill whatever career you choose

Generative and Collaborative AI

AI isn't going to replace you — but someone who knows how to work with it might. This module makes sure that's you. You'll build three essential “fusion skills” for working alongside generative AI — asking it the right questions, judging its outputs critically, and teaching it to improve — then apply them to a real project of your own design.

You'll explore:

- Prompting techniques, from chain-of-thought to few-shot learning
- Critically evaluating AI-generated outputs for bias and accuracy
- Ethical, responsible use of generative AI
- A final project building a real generative AI solution

Digital Disruption and Innovation

Got an idea that could change an industry? This module teaches you how to turn it into a business. You'll learn the techniques real tech start-ups use to launch digital products, from lean methodology and business modelling to pitching for funding, all grounded in legal and ethical best practice.

You'll explore:

- The Lean Startup method and Business Model Canvas
- Digital marketing, branding and intellectual property
- Spotting disruption through technology horizon scanning
- Pitching innovative digital solutions

Safety Critical Systems

When software failure means lives are at risk, ‘good enough’ isn't good enough — this module shows you what is. From aircraft to nuclear power stations, you'll explore the techniques used to analyse, model and mitigate risk in systems where failure simply isn't an option — including the unique safety challenges of AI and machine learning.

You'll explore:

- Fault Tree Analysis and Reliability Block Diagrams
- Markov Chains and Bayesian Networks for safety modelling
- Safety challenges unique to machine learning and artificial intelligence
- Real research applications, from drones to robotics

Secure Digital Infrastructure

Keeping the lights on for a business's digital systems is a job that never really stops — this module shows you what it takes. You'll learn to deploy, secure and maintain the infrastructure modern business runs on, from databases and email to disaster recovery.

You'll explore:

- Secure configuration of core business services
- Risk management and disaster recovery
- Systems and capacity monitoring
- Core infrastructure services — DNS, backup and virtualisation

Level 7

Commercial Development Practice

Build real software for a real client with industry-standard project management techniques and commercial tools. In Commercial Development Practice, the School's own commercial software development unit, you'll work on genuine client projects that build the business and technical skills employers tell us are actively looking for.

You'll explore:

- Requirements analysis, defining specifications and system design
- Team-based implementation and quality control
- Client liaison, negotiation and project management
- Professional, reflective practice

Component Based Software Engineering Practice

From your phone to the server room, modern software runs on portable and reusable components — learn expert-level techniques to build the platforms that make that possible. Using .NET and the .NET CLR as your guide, you'll explore component-based programming environments and virtual machines, and are design, build and secure your own component-based system.

You'll explore:

- Versioning, metadata and reflection
- Virtual machine design and architecture
- Security in virtual machine execution

- Building a virtual machine

Cutting Edge Concepts

The textbooks can't keep up with computer science — so this module looks at what's happening right now. You'll work with staff to explore genuinely current research across AI, games, security and beyond, learning to critically evaluate academic sources and separate rigorous research from the rest.

You'll explore:

- Identifying high-quality academic sources
- Critically reviewing current research
- Exploring topics beyond the core curriculum
- Skills that carry straight into postgraduate study or research careers

Dependable Reactive Real-Time Systems

Core tools: HiP-HOPS and SafeML for dependability analysis and understanding when machine-learning may no longer be trusted.

In avionics, autonomous vehicles and robotics, systems often have only one chance to get things right. This module looks at how complex software-intensive systems react and respond in real time, tolerate faults and remain safe when things go wrong.

You'll explore:

- Specification and modelling of reactive real-time systems
- Concurrency, communication and resource management
- Hazard identification and risk analysis
- Model-based dependability assessment.
- Fault tolerance, resilience and recovery

Trustworthy Computing

Security isn't an afterthought — it's a discipline. This module teaches you to build it in from the start, exploring the principles behind systems that are genuinely secure and private, from cryptography to the ethical and legal responsibilities that come with it.

You'll explore:

- Trust, authentication and security risk
- Cryptography and cryptographic algorithms
- The role of virtual machines in system security
- Securing distributed systems